KPK BOARD NOTES • ENGLISH

Dynamics

Unit 3 — Physics, Class 9

Force, Newton’s laws, momentum and friction — explained simply, with formulas, solved numericals and board-style practice on every topic.

In this unit you will learn

1. Force and inertia — why objects resist a change in motion
2. Newton’s three laws of motion, with daily-life examples
3. Mass and weight — the difference examiners love to ask
4. Momentum and the law of conservation of momentum
5. Friction — its types, advantages, disadvantages and control
6. Centripetal force — the force behind circular motion

How to use these notes

Read one section at a time. Learn the yellow formula boxes by heart, then try the numericals without looking at the steps. Finish with the MCQs and short questions on the last pages.

Board tip

Dynamics is a high-weightage unit. Almost every paper asks: one law of motion, mass vs weight, a momentum numerical, and one question on friction. Master those four and you cover most of the unit.

Based on *A Textbook of Physics, Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar. Prepared as study notes in English for Class 9 students.

Key terms at a glance

Learn these words first. Every definition below is written in board-ready language — you can use it directly in the paper.

Term	Definition
Force	A push or pull that can change the state of rest or motion of a body, change its direction, or change its shape. SI unit: newton (N).
Inertia	The property of a body by which it resists any change in its state of rest or of uniform motion. Mass is the measure of inertia.
Momentum	The quantity of motion of a moving body, equal to mass × velocity. SI unit: kg m s^{-1}.
Friction	The force that opposes motion (or the tendency of motion) between two surfaces in contact.
Centripetal force	The force that acts towards the centre of a circle and keeps a body moving in a circular path.
Weight	The force with which the Earth pulls a body towards its centre. Weight = mass × g.

Symbols and units used in this unit

Quantity	Symbol	Formula	SI unit
Force	F	F = ma	newton (N)
Acceleration	a	a = (v − u) / t	m s^{-2}
Momentum	p	p = mv	kg m s^{-1}
Weight	w	w = mg	newton (N)
Centripetal force	F_c	$F_c = mv^2 / r$	newton (N)

> **Remember:** g = 10 m s^{-2} is used in Class 9 numericals unless the question gives another value. Always write the unit with your final answer — a correct number without a unit loses marks.

1. Force and inertia

What is a force?

A **force** is a push or a pull. A force applied on a body can: (1) move a body at rest, (2) stop or slow a moving body, (3) change the speed or direction of a moving body, and (4) change the shape or size of a body, for example when clay is pressed.

Balanced and unbalanced forces

Balanced forces

The net (resultant) force is zero. The body stays at rest or keeps moving with the same velocity. Example: a book lying on a table — weight pulls it down, the table pushes it up.

Unbalanced forces

The net force is not zero. The body accelerates in the direction of the larger force. Example: kicking a football — the kick is greater than friction, so the ball speeds up.

Inertia — the laziness of matter

Inertia is the tendency of a body to keep its present state: a body at rest wants to stay at rest, and a body in motion wants to keep moving in a straight line at the same speed. A heavier body has more inertia, so mass is the measure of inertia.

Daily-life examples of inertia

- Passengers jerk **forward** when a bus stops suddenly — their bodies keep moving.
- Passengers jerk **backward** when a bus starts suddenly — their bodies try to stay at rest.
- Dust comes out of a carpet when it is beaten — the carpet moves, the dust stays.
- A coin falls into a glass when the card under it is flicked away quickly.
- It is harder to push a loaded trolley than an empty one — more mass, more inertia.

> **Think:** Inertia is not a force. It is a property of matter. Never write “force of inertia” in the paper — write “due to inertia”.

2. Newton’s laws of motion

First law — the law of inertia

A body at rest stays at rest, and a body in uniform motion in a straight line stays in that motion, unless an **external unbalanced force** acts on it. This law defines force and explains inertia.

Second law — the law of acceleration

When a net force acts on a body, it produces acceleration in the direction of the force. The acceleration is directly proportional to the force and inversely proportional to the mass of the body.

$$F = m\,a \quad | \quad a = F / m \quad | \quad 1\ \text{N} = 1\ \text{kg} \times 1\ \text{m s}^{-2}$$

Example: The same force gives a light football far more acceleration than a heavy roller. A stronger engine gives a car greater acceleration.

Third law — action and reaction

To every action there is an **equal and opposite reaction**. Action and reaction act on *different* bodies, so they never cancel each other.

- **Walking:** your foot pushes the ground backward (action); the ground pushes you forward (reaction).
- **Recoil of a gun:** the bullet moves forward; the gun moves backward.
- **Rocket:** hot gases are pushed down; the rocket is pushed up.
- **Swimming and rowing:** water is pushed backward; the swimmer or boat moves forward.
- **Bird flight:** wings push air down and back; air pushes the bird up and forward.

> **Common mistake:** “Action and reaction cancel, so nothing should move.” Wrong — they act on different bodies, so each body feels only one of them.

3. Mass and weight

Point	Mass	Weight
Meaning	Quantity of matter in a body	Force with which Earth pulls the body
Symbol / formula	m	w = mg
SI unit	kilogram (kg)	newton (N)
Type	Scalar quantity	Vector quantity (towards Earth's centre)
Change of place	Remains the same everywhere	Changes with g — less on the Moon, more on Jupiter
Measured by	Beam / physical balance	Spring balance
Can it be zero?	Never zero	Zero in free fall / deep space (weightlessness)

w = m g — a 5 kg body weighs 5 × 10 = 50 N on Earth

4. Momentum

Momentum is the quantity of motion of a body. It depends on both mass and velocity — a slow truck can have more momentum than a fast bicycle.

p = m v

- Momentum is a **vector** quantity; its direction is the direction of velocity.
- SI unit: **kg m s^{-1}** (also written N s).
- A body at rest has zero momentum because v = 0.
- **Rate of change of momentum** equals the applied force: F = (mv − mu) / t. This is how Newton's second law is also stated.

> **Why seat belts and airbags work:** In a crash the momentum changes by the same amount, but a belt stretches the stopping time t. Longer time means smaller force (F = change in momentum / t), so injury is reduced. The same idea explains why a cricketer pulls his hands back while catching a ball.

Law of conservation of momentum

Statement: When two or more bodies collide and no external force acts on them, the total momentum before collision is equal to the total momentum after collision. Momentum is never created or destroyed in the collision — it is only transferred.

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

How the law is proved (in words)

1. Two balls of masses m_1 and m_2 move with velocities u_1 and u_2 and collide for a short time t.
2. By Newton's third law, the force of ball 1 on ball 2 is equal and opposite to the force of ball 2 on ball 1.
3. So the change of momentum of one ball is equal and opposite to the change of momentum of the other.
4. Therefore total momentum after collision = total momentum before collision.

If the bodies join after collision

$$m_1u_1 + m_2u_2 = (m_1 + m_2)\, v$$

Daily-life applications

- **Recoil of a gun:** bullet momentum forward = gun momentum backward. A heavier gun recoils less.
- **Rocket and jet:** exhaust gases gain momentum one way; the rocket gains equal momentum the other way.
- **Explosion:** pieces fly in different directions so that total momentum stays zero if it was zero at first.
- **Collision of vehicles:** investigators use this law to find speeds before a crash.

> **Direction matters:** Take motion to the right as positive and to the left as negative. Most errors in momentum numericals are sign errors, not formula errors.

5. Friction

Friction is the force that opposes motion between two surfaces in contact. It always acts opposite to the direction of motion (or attempted motion). It is caused by the roughness and interlocking of the two surfaces — even polished surfaces are rough under a microscope.

Types of friction

Type	When it acts	Size
Static friction	Body is at rest; force is applied but body does not move yet	Adjusts itself up to a maximum — the **limiting friction**
Kinetic (sliding) friction	Body is sliding over a surface	Slightly less than limiting friction
Rolling friction	Body rolls, e.g. wheel or ball	Much smaller than sliding friction

Friction $f = \mu R$ (μ = coefficient of friction, R = normal reaction)

Friction increases with roughness and with the force pressing the surfaces together (the weight). It does not depend on the area of contact (at Class 9 level).

Advantages

- Walking and running are possible
- Vehicles grip the road; brakes work
- Nails, screws and knots hold
- Writing with pen or chalk
- Lighting a match

Disadvantages

- Wears out tyres, shoes and machine parts
- Wastes energy as heat
- Reduces speed and efficiency of machines
- Produces unwanted noise

How friction is reduced

- **Lubrication:** oil or grease forms a smooth layer between parts.
- **Ball bearings:** change sliding friction into much smaller rolling friction.
- **Polishing** surfaces and **streamlining** vehicles, boats and aircraft.

How friction is increased when needed

Grooves (treads) on tyres, spikes on athletes' shoes, rough brake pads, and sand thrown on icy roads.

6. Centripetal force

When a body moves in a circle with uniform speed, its direction changes at every point. A changing direction means a changing velocity, so an acceleration — and therefore a force — must act on it. This force always points **towards the centre** of the circle.

$$F_c = m\,v^2 / r \quad \text{and} \quad a_c = v^2 / r$$

- F_c increases if mass or speed increases (speed matters most — it is squared).
- F_c decreases if the radius of the circle increases — wider turns need less force.

What provides the centripetal force?

Situation	Centripetal force is provided by
Stone whirled on a string	Tension in the string
Car turning on a flat road	Friction between tyres and road
Moon around the Earth	Gravitational pull of the Earth
Electron around the nucleus	Electrostatic attraction
Cyclist leaning into a turn	Component of weight and normal reaction

Centrifugal effect

If the string breaks, the stone does not fly outward — it flies off along the **tangent**, because the centre-seeking force has disappeared. The “outward throw” you feel in a turning car is your body’s inertia trying to keep it moving straight. Washing-machine dryers and cream separators use this effect.

> **Banked roads:** Curves on highways and velodromes are tilted (banked) so a component of the normal reaction helps provide the centripetal force. Less friction is then needed and fast turns become safer.

At a glance — all Unit 3 formulas

$F = ma$	$p = mv$	$w = mg$
$F = (mv - mu) / t$	$f = \mu R$	$F_c = mv^2 / r$

Solved numericals

Method for every numerical: write Given, write the formula, put in values with units, calculate, box the answer with its unit.

Numerical 1 — Newton's second law

Q: A force of 100 N acts on a body of mass 20 kg. Find its acceleration.

Solution: $F = ma \rightarrow a = F / m = 100 / 20 =$ **$5\ m\ s^{-2}$**.

Numerical 2 — Momentum

Q: Find the momentum of a 1200 kg car moving at 15 m s^{-1}.

Solution: $p = mv = 1200 \times 15 =$ **$18{,}000\ kg\ m\ s^{-1}$**.

Numerical 3 — Weight

Q: A student has mass 45 kg. Find his weight on Earth ($g = 10\ m\ s^{-2}$) and on the Moon ($g = 1.6\ m\ s^{-2}$).
Solution: Earth: $w = 45 \times 10 =$ **450 N**. Moon: $w = 45 \times 1.6 =$ **72 N**. Mass stays 45 kg in both places.

Numerical 4 — Conservation of momentum

Q: A 2 kg ball moving at 6 m s^{-1} collides with a stationary 1 kg ball and both move together. Find their common velocity.

Solution: $m_1u_1 + m_2u_2 = (m_1 + m_2)v \rightarrow (2 \times 6) + (1 \times 0) = 3v \rightarrow v = 12 / 3 =$ **$4\ m\ s^{-1}$**.

Numerical 5 — Recoil of a gun

Q: A bullet of mass 0.02 kg leaves a 4 kg gun at 200 m s^{-1}. Find the recoil velocity of the gun.

Solution: Total momentum before = 0, so $0 = (0.02 \times 200) + (4 \times v) \rightarrow 4v = -4 \rightarrow v =$ **$-1\ m\ s^{-1}$** (minus sign = opposite direction).

Numerical 6 — Centripetal force

Q: A 0.5 kg stone is whirled in a circle of radius 2 m at 4 m s^{-1}. Find the centripetal force.

Solution: $F_c = mv^2 / r = (0.5 \times 16) / 2 =$ **4 N**.

Practice — MCQs

1. SI unit of force is: (a) kg m s^{-1} (b) **newton** (c) joule (d) watt

2. Inertia of a body depends on its: (a) velocity (b) **mass** (c) shape (d) volume

3. Momentum of a 2 kg body moving at 5 m s^{-1} is: (a) 2.5 (b) 7 (c) **10 kg m s^{-1}** (d) 25

4. Action and reaction act on: (a) same body (b) **different bodies** (c) nothing (d) only solids

5. Which friction is smallest? (a) static (b) sliding (c) **rolling** (d) limiting

6. Centripetal force acts: (a) outward (b) along tangent (c) **towards the centre** (d) downward only

Short questions for practice

1. State Newton's second law and derive F = ma in your own words.
2. Why do passengers fall forward when a vehicle stops suddenly?
3. Give three differences between mass and weight.
4. State the law of conservation of momentum and give two applications.
5. Write three advantages and three disadvantages of friction.
6. How is friction reduced in machines? Give three methods.
7. Why is a cyclist forced to lean inward while taking a turn?
8. A fielder pulls his hands back while catching a fast ball. Why?

One-page revision checklist

- [ ] I can define force, inertia, momentum, friction and centripetal force.
- [ ] I can state all three laws with one example each.
- [ ] I can solve F = ma, p = mv, w = mg and $F_c = mv^2/r$ numericals.
- [ ] I can prove and apply the law of conservation of momentum.

> **Prepared by Shah Fahad** — Primary School Teacher, Buner, Khyber Pakhtunkhwa. For more free notes, visit the notes library on the website. Contact: **03409090945**.